

4 Coordinate systems	4.1	Cartesian and parametric equations for the parabola and rectangular hyperbola , ellipse and hyperbola.	Students should be familiar with the equations: $y^2 = 4ax; \quad x = at^2, \quad y = 2at$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1; \quad x = a \cos t, \quad y = b \sin t.$ $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1; \quad x = a \sec t, \quad y = b \tan t,$ $x = a \cosh t, \quad y = b \sinh t$
	4.2	The focus-directrix properties of the parabola , ellipse and hyperbola, including the eccentricity.	For example, students should know that, for the ellipse $b^2 = a^2(1 - e^2)$, the foci are $(ae, 0)$ and $(-ae, 0)$ and the equations of the directrices are $x = +\frac{a}{e} \quad \text{and} \quad x = -\frac{a}{e}$
	4.3	Tangents and normals to these curves.	The condition for $y = mx + c$ to be a tangent to these curves is expected to be known. Students are expected to be able to use algebraic differentiation to find $\frac{dy}{dx}$ for these curves.
	4.4	Loci problems.	